2026 BROCK AI DAY · 3-MINUTE RESEARCH PRESENTATION

Are language models converging on shared structures?

Do AI models built by different teams, from different text, organize concepts the same way?

James R. A. Henry · Computer Science, Brock University

BACKGROUND

How to look inside a language model

1

Mechanistic interpretability

Study a model by reading its internal activity, not just its answers. That needs the model's weights, so we use open models.

2

Concepts are directions

A model stores each concept it learns, like *negation* or *credibility*, as a direction inside it. The Linear Representation Hypothesis.

3

One world, one map?

Models trained on different data may converge on the same picture of the world. The Platonic Representation Hypothesis.

Park, Choe & Veitch (2024), *The Linear Representation Hypothesis and the Geometry of Large Language Models*, ICML · Huh, Cheung, Wang & Isola (2024), *The Platonic Representation Hypothesis*, ICML.

PROBLEM

Do different models share one map of concepts?

Every model stores concepts in its own private coordinates, so you cannot lay one model's directions over another's.

But the **relationships** between concepts could still match: is credibility close to formality and opposed to sarcasm in every model?

WHY IT MATTERS

If the map is shared, what we learn inside one model can be checked in others.

And it tells us whether models learn something about the world, or just quirks of their training.

METHOD

Find each concept, then compare the angles

1

Write pairs

The same passage twice, with and without the concept. 17 concepts, 250 pairs each.

2

Find the directions

Run both through 35 open models from 10 families. The difference is the concept's direction.

3

Compare angles

Inside each model, measure how close each concept is to every other. Angles can be compared across models; coordinates cannot.

4

Try to fail

Hide a whole model family. Can the other families' angles alone identify its concepts?

Models include Llama, Mistral, Qwen, Gemma, Pythia, OPT and GPT-2, from 70 million to 72 billion parameters.

RESULTS

The hidden family's concepts are identified

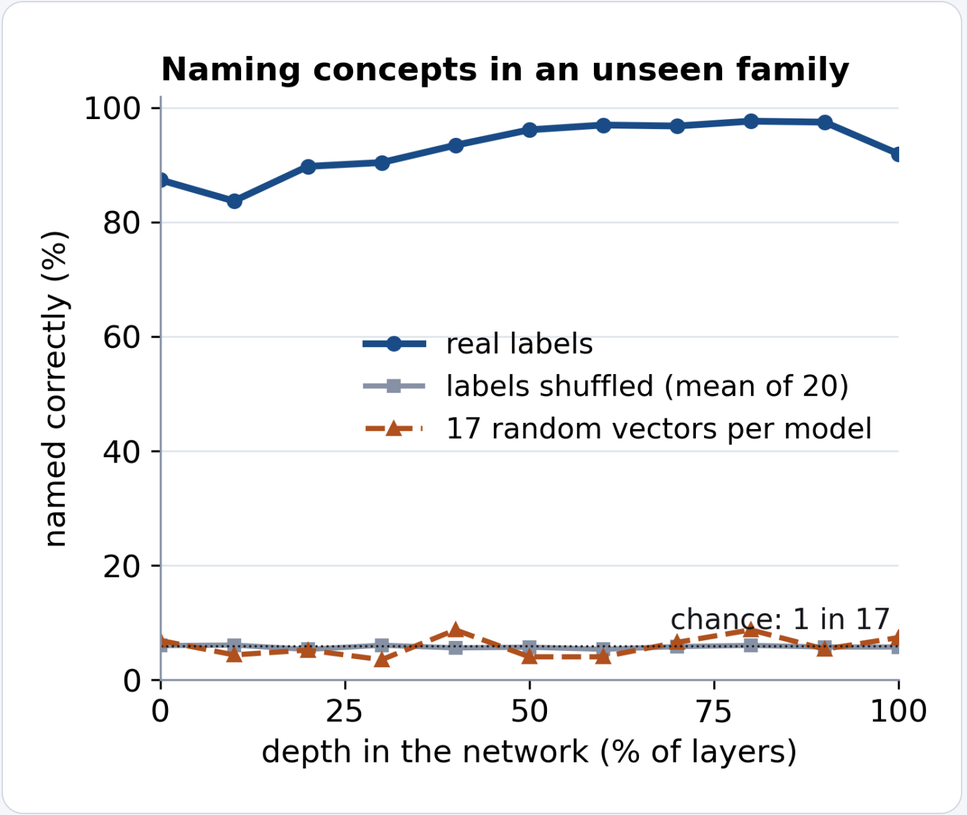
97%

of concepts identified correctly in a hidden model family, in the middle layers

6%

with the labels shuffled, or with random directions: chance

One example: in 29 of 35 models, the concept most closely tied to credibility is formality.



CONCLUSION

Different models, one map of concepts

Language models built by different teams, from different text, organize concepts the same way. That supports the Platonic Representation Hypothesis, measured concept by concept.

Because the map is shared, findings from inside one model can be tested on others.

Is this the structure of language itself, or of the world it describes?

Come and see the poster. Code, data and reading list: github.com/jamesrahenry/ai-day-2026